THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
033 BIOLOGY
MARKING SCHEME NOVEMBER 2025

SECTION A (15 MARKS)

1. **(10 marks)**

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	A	B	B	D	C	D	C	D	A

2. **(05 marks)**

(i)	(ii)	(iii)	(iv)	(v)
G	A	E	F	C

SECTION B (70 MARKS)

3. (a)Biology:
Biology is the scientific study of life and living organisms, including their structure, function, growth, origin, evolution, and distribution. **(02 Marks)**
(b) Importance of studying Biology **(any four @02 Marks)**
   (i) Understanding of self: It helps us understand our bodies, how they function, and how to maintain good health.
   (ii) Health and disease: It provides knowledge on the causes, transmission, and prevention of diseases, contributing to the development of vaccines and treatments.
   (iii) Environmental conservation: It provides insight into the interactions between organisms and their environment, promoting sustainable management of natural resources.
   (iv) Agriculture and food security: It is vital for improving crop yields, developing disease-resistant plant varieties, and enhancing livestock production to ensure sufficient food supply.
   (v) Technological advancements**:** It contributes to breakthroughs in biotechnology, genetics, and medicine, leading to new diagnostic tools and treatments.

Total: 10 Marks

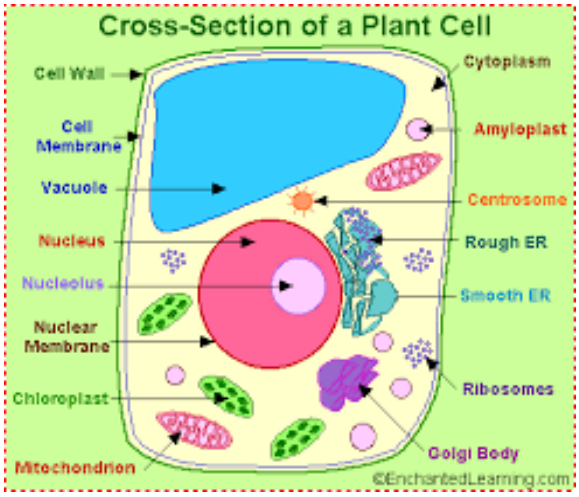
4. (a) Similarities between plant and animal cells: **(any three @02 Marks)**
   (i) **Cytoplasm:** Both cells have cytoplasm, a jelly-like substance where metabolic reactions occur.

(ii) **Cell membrane:** Both are enclosed by a cell membrane that regulates the movement of substances.

(iii) **Nucleus:** Both possess a nucleus, which controls cell activities and contains genetic material.

(iv) **Mitochondria:** Both contain mitochondria, where cellular respiration takes place to generate energy.

(b) Diagram of a plant cell:



Caption: 00½ mark
Diagram: 01 mark
Labeling: 5 × ½ = 02½ marks

5. **(a) Binomial nomenclature: (02 Marks)**
Binomial nomenclature is a formal system of naming species whereby each species is given a name consisting of two parts. The first part is the genus, and the second part is the species name (e.g., *Homo sapiens* for humans).

(b) Characteristics of the kingdom Monera: (any four @02 Marks)

(i) **Unicellular:** All organisms in this kingdom are single-celled.

(ii) **Prokaryotic:** Their cells lack a membrane-bound nucleus and other membrane-bound organelles.

(iii) **Microscopic:** They are very small and can only be observed with a microscope.

(iv) **Reproduce asexually:** They reproduce mainly through binary fission.

6. Mushroom: Basidiomycota, Yeast: Ascomycota, and Mucor: Zygomycota **(01 @ = 03 marks)**
Features: Cell wall made of chitin, Saprophytic… (01 mark)

(b) Advantages of the Kingdom Fungi are:

(i) They help in the decomposition of organic matter and thus making the soil fertile

(ii) Some mushrooms are used as food and source of income

(iii) Some fungi are used in bread making by causing the dough to rise, example, yeasts

(iv) Some fungi are used in the production of alcohol, example, yeasts

(v) Some fungi are used in the manufacturing of medicine, example penicillium
(vi) Some fungi are used in scientific researches, example, researches in genetic engineering

(Any four: 01½ @ =06 *marks)*

(Total: 10 marks)

7. **(02 @ =10 marks)**
   (i) Anatomy: the study of the structure of plant and animal organs.
   (ii) Histology: the study of the structure of tissues.
   (iii) Microbiology: the study of microscopic organisms.
   (iv) Bacteriology: the Study of bacteria.
   (v) Virology: the study of viruses.

(Total: 10 marks)

8. (a)(i) Photosynthesis is the process whereby green plants make carbohydrate by using water, carbon dioxide and the sunlight. (**02 marks)**
   (ii) The green plants use chlorophyll to trap energy from the sunlight and uses it combine water and carbon dioxide. (**02 marks)**
   (b) Photosynthesis
   (ii) Help to manufacture food
   (iii) Help to reduce carbon dioxide in the atmosphere
   (iv) Help to add oxygen in the atmosphere
   (v) Help to convert sunlight energy to chemical energy **01½ marks @ = (06 marks)**

9. Features of a good Biology laboratory **(Any ten @1 = 10 marks)**
   (i) Should have enough supply of air/must be well ventilated.
   (ii) Should have electricity.
   (iii) Should have enough water supplies.
   (iv) Should have fire extinguishers.
   (v) Should have preserved specimens.
   (vi) Should have models.
   (vii) Should have oven for drying samples.
   (viii) Should have enough space.
   (ix) Should have enough heat sources.
   (x) Should have refrigerator or freezer for storing of specimen.
   (xi) Should have non-slippery floor.
   (xii) Should have proper lighting.

(Total: 10 marks)

SECTION C (15 MARKS)

10. Importance of Bacteria

Introduction 1 1/2 Mark

Importance of bacteria

- Bacteria in ruminant gut manufacture vitamin K and B12
- They are used in decay and decomposition

- Enhancing soil fertility
- They are used in genetic engineering Any 6 @2 marks (Total 12 marks)
- They are used in manufacturing of vaccine
- They are used in biological study and research
- They serve as biological weapons
- Autotrophic bacteria carry out photosynthesis

Conclusion 1 1/2 mark

Total: 15 Marks